

LOVELY PROFESSIONAL UNIVERSITY

Academic Task No. 1

School: SCA

Faculty of: LFTS

Course Title: PROGRAMMING IN JAVA

Course Code: CAP615

Max. Marks: 30

Date of Allotment: 14th Feb. 2022

Last Date of Submission: 22nd Feb. 2022

Important Guidelines:

1. All questions in this Academic Task are compulsory.
 2. It is mandatory to attempt all questions of the assignment in your own handwriting on A4 size sheets/pages with a blue color ink pen. Any other mode of attempt (typed or printed codes or table) except hand written/drawn will not be accepted/considered as valid submission(s) under any circumstances.
 3. Every attempted sheet/page should carry clear details of student such as Name, Registration number, Roll number, Question number and Page number. The page numbers should be written clearly on the bottom of every attempted sheet in a prescribed format as: for page 1; **Page 1 of 4**, for page 2; **Page 2 of 4**, for page 3; **Page 3 of 4** and for page 4; **Page 4 of 4**, in case your assignment/document is of 4 pages.
 4. After attempting the answer(s), student needs to take photograph of each of these answer sheets/pages and needs to convert the jpeg format images into a sequential single pdf format document (can be done with many free online available converters).
 5. This PDF file should be uploaded onto the UMS interface on or before the last date of the submission.
 6. Refrain from indulging into plagiarism as copy cases will be marked zero
 7. For every program it is mandatory to give proper documentation.
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SET-1

Q1: Define any five features of Java Language and compare with C, C++ programming language. [10 Marks]

Q2: Define Local variable, Static variable and the Instance variable with the help of one example. [5 Marks]

Q3: Define public static void main (String args []) in Java. [5 Marks]

Q4. Analyze the difference between rectangular and jagged array in Java [10 Marks]

SET-2

Q1: Java is object-oriented programming define object-oriented programming features.

[10 Marks]

Q2: Suppose you have string "hello". How many ways you can create this string object? Analyze with code.

[5 Marks]

Q3: With the help of example define Autoboxing and Unboxing feature in Java.

[5 Marks]

Q4: When writing programs, programmers may commit few common mistakes while using the if- else statements. Analyze these mistakes with the help of example.

[10 Marks]

SET-3

Q1: Analyze the difference between for loop and advance for loop. How advance for loop is used in 2D array? write one example.

[10 Marks]

Q2: With the help of example define Encapsulation features in Java.

[10 Marks]

Q3: Define and differentiate the following pairs:

(a) the logical && and || operators

(b) a break and a continue statement

[10 Marks]

SET-4

Q1: "Various string methods are used for different tasks of string manipulation" Define these methods with the help of example.

[10 Marks]

Q2: In which type of application, we can use the 3-D array? Analyze with an example.

[10 Marks]

Q3: Define and differentiate the following pairs:

(a) the bitwise >> and << operators

(b) a while statement and a do while statement

[10 Marks]

SET-5

Q1: Define how a string class object can be created using an existing 'String Buffer' object? Also, define how can you find the location of the last occurrence of 'a' in the string "Java Programming"?

[10 Marks]

Q2: Define different scopes of variables in Java.

[5 Marks]

Q3: Define the concept of String pool in Java.

[5 Marks]

Q4: Which type of variables used in a switch statement? Analyze those variables with the help of example.

[10 Marks]

SET-6

Q1: Define any five features of Java Language and compare with C, C++ programming language.

[10 Marks]

Q2: When writing programs, programmers may commit few common mistakes while using the if- else statements. Analyze these mistakes with the help of example.

[10 Marks]

Q3: "2-dimensional arrays can be created in two ways" Justify your answer with the help of example.

[10 Marks]

SET-7

Q1: Analyze the difference between for loop and advance for loop. How advance for loop is used in 2D array? write one example.

[10 Marks]

Q2: "Various string methods are used for different tasks of string manipulation" Define these methods with the help of example.

[10 Marks]

Q3: With the help of example define Encapsulation features in Java.

[10 Marks]

SET-8

Q1: Define how a string class object can be created using an existing 'String Buffer' object? Also, define how can you find the location of the last occurrence of 'a' in the string "Java Programming"?

[10 Marks]

Q2: In which type of application, we can use the 3-D array? Analyze with an example.

[10 Marks]

Q3: Define different scopes of variables in Java.

[5 Marks]

Q4: Define the concept of String pool in Java.

[5 Marks]

SET-9

Q1: Define and differentiate the following pairs:

(a) the bitwise >> and << operators

(b) a while statement and a do while statement

[10 Marks]

Q2: Which type of variables used in a switch statement? Analyze those variables with the help of example. [10 Marks]

Q3: "2-dimensional arrays can be created in two ways" Justify your answer with the help of example. [10 Marks]

List of Students:

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5	12113075	Ankit Kumar Mishra	RD2110A07	Set-5
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7	12110164	Vishal Ojha	RD2110A10	Set-7
8	12110187	Renbithung M Ngullie	RD2110A11	Set-8
9	12110286	Himansu Kumar Das	RD2110A12	Set-9
10	12110287	Manish Kumar Tripathy	RD2110A13	Set-1
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14	12106059	Sachin Pal	RD2110A17	Set-5
15	12108431	Kajal Kumari	RD2110A18	Set-6
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17	12105973	Harsh Sood	RD2110A21	Set-8
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27	12101666	Monika Rastogi	RD2110A31	Set-9
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35	12100232	Satish Kumar Choudhary	RD2110A39	Set-8
36	12100272	Gulam Nabi	RD2110A40	Set-9
37	12100268	Avinash Kumar	RD2110A41	Set-1
38	12100165	Dolly Kumari	RD2110A42	Set-2
39	12100139	Manish Das	RD2110A43	Set-3
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51	12102541	Sonia	RD2110B55	Set-6
52	12102530	Aryan Mot	RD2110B56	Set-7
53	12102924	Ajay Upadhyay	RD2110B57	Set-8
54	12102983	Atul Kumar Gupta	RD2110B58	Set-9
55	12102947	Ishika Keshwani	RD2110B59	Set-1
56	12103005	Raju Basak	RD2110B60	Set-2
57	12102850	Abhinav Kumar	RD2110B62	Set-3
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59	12102875	Trina Mahapatra	RD2110B64	Set-5
60	12102895	Nishant Raj Singh	RD2110B65	Set-6
61	12102899	Nitin Shukla	RD2110B66	Set-7
62	12102200	Abhinay Sharma	RD2110B67	Set-8
63	12102231	Deepak Kumar	RD2110B68	Set-9
64	12102235	Nitesh Khatri	RD2110B69	Set-1
65	12102245	Ashwani Kumar	RD2110B70	Set-2
66	12102284	Rohan Sharma	RD2110B71	Set-3
67	12102282	Thota Venkata Sai	RD2110B72	Set-4
68	12102289	Vivek Kumar	RD2110B73	Set-5
69	12102293	Aadil Ahmad Yatoo	RD2110B74	Set-6
70	12102130	Amit Babani	RD2110B75	Set-7
71	12102134	Akhil Kumar Verma	RD2110B76	Set-8
72	12102137	Saksham Verma	RD2110B77	Set-9
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86	12108912	Aakash Kalra	RD2110B91	Set-5
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89	12110972	Gaurav Kumar	RD2110B94	Set-8
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